

Notes: Brennan and Subrahmanyam (JFE, 1995)

Investment Analysis and Price Formation in Securities Markets

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Economic mechanism	3
3.1	Why analyst following should matter	3
3.2	Core intuition	4
4	Model background	4
4.1	Kyle price impact	4
4.2	Admati-Pfleiderer prediction	4
5	Data	4
6	Measurement of adverse-selection cost	5
6.1	Glosten-Harris specification	5
6.2	Madhavan-Smidt specification	5
6.3	Foster-Viswanathan robustness specification	5
7	Empirical design	5
8	Main findings	6
8.1	Reduced-form evidence	6
8.2	Structural evidence	6
9	Extracted tables from the paper	6
9.1	Table 1: Summary statistics of estimated market depth and analyst following	6
9.2	Table 2: Two-stage least-squares estimates of determinants of adverse-selection cost	7
9.3	Table 3: Robustness using the Foster-Viswanathan market-depth measure	8
9.4	Table 4: Structural Admati-Pfleiderer model using trading-volume volatility	9
9.5	Table 5: Structural Admati-Pfleiderer model using firm size as noise-trading proxy .	10
9.6	Table 6: Structural robustness with institutional ownership variables	11
10	Interpretation and implications	11
10.1	For market microstructure	11
10.2	For analyst coverage	12
10.3	For asset pricing	12
11	Limitations	12
12	Connection to other papers	12
13	Short summary	12

1 Big picture

This note summarizes Brennan and Subrahmanyam (1995), which studies whether security analysis affects market liquidity. The paper asks a precise market microstructure question: after controlling for volume, price, volatility, size, and institutional ownership, do stocks followed by more analysts have lower adverse-selection trading costs?

The main answer is yes. Using intraday NYSE data from 1988, the authors estimate adverse-selection costs using Glosten-Harris, Madhavan-Smidt, and Foster-Viswanathan style procedures. Across specifications, more analyst coverage is associated with lower inverse market depth and therefore deeper markets. The economic interpretation is that more analysts increase competition among informed agents and accelerate information production, reducing the informational disadvantage faced by market makers.

2 Incremental contribution of the paper

The paper contributes in four incremental ways.

1. **It moves the microstructure literature from measurement to cross-sectional explanation.** Earlier work, such as Glosten and Harris (1988), estimated adverse-selection components of bid-ask spreads, but provided limited evidence on what explains cross-sectional differences in those costs. This paper uses analyst following as a direct cross-sectional proxy for information production.
2. **It links information production to liquidity.** The key idea is not merely that informed trading creates adverse selection, but that competition among informed agents can make markets deeper by making information impounded into prices more efficiently.
3. **It empirically connects analyst coverage to Kyle-type market depth.** The paper defines adverse-selection cost as the price impact of a marginal dollar of trade, closely related to the inverse of Kyle market depth.
4. **It structurally tests a version of the Admati-Pfleiderer model.** Beyond reduced-form regressions, the paper estimates nonlinear simultaneous equations implied by endogenous information acquisition and finds qualified support.

3 Economic mechanism

3.1 Why analyst following should matter

The paper treats the number of analysts following a firm as a proxy for the amount of information production about that firm. A stock with more analyst coverage has more agents generating, processing, and disseminating information. The theoretical prediction is subtle:

- In Kyle-style settings, informed trading creates adverse-selection costs for liquidity suppliers.
- In Admati and Pfleiderer (1988), when multiple informed traders compete, greater competition can increase market depth.
- With long-lived information, as in Holden and Subrahmanyam (1992), more informed traders may reveal information faster, making market depth time-varying but higher on average.

3.2 Core intuition

Market makers protect themselves against informed order flow by adjusting prices more aggressively to trades. If a trade is more likely to contain private information, the price impact of order flow is larger. However, when there are many informed agents, each agent's informational rents are competed away and information is reflected in price more quickly. This competition can reduce the marginal adverse-selection cost faced by liquidity traders.

4 Model background

4.1 Kyle price impact

The market maker sets price as a linear function of order flow:

$$P = E[u] + \lambda q, \quad (1)$$

where u is the terminal payoff and q is total order flow. The parameter λ is the inverse of market depth: a larger λ means a larger price movement for a given order flow and therefore worse liquidity.

4.2 Admati-Pfleiderer prediction

In a special case with n risk-neutral informed traders who receive a perfectly informative signal, the inverse of depth is

$$\lambda = \frac{\sqrt{n}}{n+1} \sqrt{\frac{\text{var}(u)}{\text{var}(z)}}, \quad (2)$$

where z is noise-trader order flow. For $n > 1$, λ decreases as n rises. Therefore, more informed traders imply a deeper market.

If information acquisition is costly, the equilibrium number of informed traders satisfies

$$\frac{\text{var}(u)}{(1+n)^2 \lambda} = c, \quad (3)$$

where c is the fixed cost of becoming informed. This equilibrium condition implies that analyst following is endogenous: more analysts follow a stock when the private benefits of information production are large relative to costs.

5 Data

The empirical sample uses intraday quotes, transaction prices, and quantities for NYSE common stocks continuously listed during calendar year 1988. After filtering and requiring all variables, the final sample contains 1,421 stocks.

- **Intraday trades and quotes:** from the Institute for the Study of Security Markets.
- **Analyst following:** number of analysts making an annual earnings forecast in December 1987 according to I/B/E/S.
- **Trading volume, price, and return variance:** from CRSP NYSE/AMEX daily data for 1988.
- **Institutional ownership:** S&P Security Owners' Stock Guide, measured as of December 1987.

The authors screen the intraday data by removing out-of-sequence trades, overnight price changes, closing quotes, and likely reporting errors. This matters because microstructure estimates are sensitive to trade classification and price recording errors.

6 Measurement of adverse-selection cost

6.1 Glosten-Harris specification

Let D_t be the trade sign: +1 for buyer-initiated trades and -1 for seller-initiated trades. The Glosten-Harris specification estimates

$$\Delta p_t = \lambda q_t + \psi(D_t - D_{t-1}) + y_t. \quad (4)$$

Here λ captures the permanent price impact of signed order flow, interpreted as the adverse-selection component; ψ captures the fixed-cost component of the spread.

6.2 Madhavan-Smidt specification

The Madhavan-Smidt model allows informed investors to condition their order flow on price. The estimated equation is

$$\Delta p_t = \lambda q_t + \frac{\psi}{\pi} D_t - \psi D_{t-1} + \eta_t, \quad (5)$$

where the error is assumed to follow an MA(1) process.

6.3 Foster-Viswanathan robustness specification

The Foster-Viswanathan measure estimates price impact using the unexpected component of order flow. The first stage predicts order flow from lagged order flows and prices; the second stage regresses price changes on the residual order flow and trade-sign changes. This checks whether the main result depends on predictable order-flow components.

7 Empirical design

The central reduced-form system is a simultaneous equation model. The adverse-selection cost equation is

$$LTC = a_{S0} + a_{S1}LANAL + a_{S2}LVOL + a_{S3}LPRI + a_{S4}LVAR + e_{TC}, \quad (6)$$

where $LTC = \log(\lambda/PRI)$, $LANAL = \log(1 + \text{number of analysts})$, $LVOL$ is log trading volume, $LPRI$ is log price, and $LVAR$ is log return variance.

The analyst-following equation allows analyst coverage to depend on trading costs, return variance, firm size, price, industry dummies, and institutional ownership. The trading-volume equation allows volume to depend on trading cost, analyst following, and firm size. This simultaneous setup is important because analyst following and volume are themselves endogenous to liquidity.

8 Main findings

8.1 Reduced-form evidence

Across the Glosten-Harris, Madhavan-Smidt, and Foster-Viswanathan measures, the coefficient on analyst following is negative and statistically significant in the adverse-selection cost equation. This means that more analysts are associated with lower price impact, controlling for other known liquidity determinants.

The other coefficients are also economically sensible:

- Higher trading volume reduces adverse-selection costs.
- Higher return variance increases adverse-selection costs.
- Higher-priced stocks are more liquid for a given dollar transaction size.
- Analyst following is positively related to trading volume, consistent with analysts generating investor attention and brokerage activity.

8.2 Structural evidence

The structural Admati-Pfleiderer estimates broadly support the model. The coefficients generally have the predicted signs: adverse-selection cost increases with return volatility and decreases with analyst following or with proxies for noise trading. The model performs better when enriched with institutional ownership variables, suggesting that the simple theory captures the core mechanism but not all features of the market for information.

9 Extracted tables from the paper

9.1 Table 1: Summary statistics of estimated market depth and analyst following

Table 1 Summary statistics of estimated market depth for the year 1988 and analyst following as of December 1987 for a sample of 1,421 stocks continuously listed on the NYSE for the year 1988 for which complete data were available on institutional holdings as of December 1987 and for which nonnegative estimates of market depth were obtained λ_{GH} is the estimate of the inverse of market depth from the Glosten-Harris specification: $\Delta p_t = \lambda q_t + \psi [D_t - D_{t-1}] + y_t;$ λ_{MS} is the estimate of the inverse of market depth from the Madhavan-Smidt specification: $\Delta p_t = \lambda q_t + \frac{\psi}{\pi} D_t - \psi D_{t-1} + \eta_t.$ Δp_t is the price change at transaction t, q_t is the signed trade size, D_t is a dummy variable that is equal to +1 for a trade classified as a buy and -1 for a sell, and y_t and η_t are error terms. The Glosten-Harris model is estimated assuming that y_t is i.i.d., while the Madhavan-Smidt model is estimated assuming that η_t is MA(1). <i>PRI</i> is the average daily closing price for 1988.			
Number of firms = 1,421	Mean	Median	Standard deviation

Table 1: Summary statistics of estimated market depth and analyst following

Read:

- The final sample has 1,421 NYSE firms.

- The mean marginal cost of purchasing 500 shares is 3.14% using the Glosten-Harris measure and 1.56% using the Madhavan-Smidt measure.

- Analyst following is highly skewed: the mean number of analysts is 8.90, while the median is 5.

Economic message: Adverse-selection costs differ substantially across firms, and analyst coverage is unevenly distributed. This creates meaningful cross-sectional variation for testing whether information production is linked to market depth.

9.2 Table 2: Two-stage least-squares estimates of determinants of adverse-selection cost

Table 2

Two-stage least-squares estimates of determinants of adverse selection cost of transacting, using two empirical measures of λ :

$$LTC = a_{00} + a_{01}LANAL + a_{02}LVOL + a_{03}LPRI + a_{04}LVAR + \epsilon_{TC}.$$

$$LANAL = a_{10} + a_{11}LTC + a_{12}LVAR + a_{13}LSIZE + a_{14}LPRI + \sum_{i=1}^5 a_{1i+1}IND_i + a_{110}LNINST + a_{111}LPINST + \epsilon_{ANAL},$$

$$LVOL = a_{20} + a_{21}LTC + a_{22}LANAL + a_{23}LSIZE + \epsilon_{VOL}.$$

The equation for $LANAL$ is not identified.

$LTC_{GH} = \log(\lambda_{GH}/PRI)$ and $LTC_{MS} = \log(\lambda_{MS}/PRI)$ are the logs of the adverse selection costs of transacting. PRI is the average daily closing price.

λ_{GH} is the estimate of the inverse of market depth from the Glosten-Harris specification:

$$\Delta p_t = \lambda_{GH} + \psi[D_t - D_{t-1}] + y_t;$$

λ_{MS} is the estimate of the inverse of market depth from the Madhavan-Smidt model specification:

$$\Delta p_t = \lambda_{MS} + \frac{\psi}{\pi} D_t - \psi D_{t-1} + \eta_t.$$

Δp_t is the price change at transaction t , q_t is the signed trade size, D_t is a dummy variable that is equal to +1 for a trade classified as a buy and -1 for a sell, and y_t and η_t are error terms. The Glosten-Harris model is estimated assuming that y_t is i.i.d., while the Madhavan-Smidt model is estimated assuming that η_t is MA(1). The sample consists of 1,421 stocks continuously listed on the NYSE for the year 1988 for which complete data were available on institutional holdings as of December 1987 and for which nonnegative estimates of market depth were obtained.

The other variables are defined as follows: $LANAL$ is the logarithm of one plus the number of analysts as of December 1987, $LVOL$ is the logarithm of the average daily trading volume in 1988, $LPRI$ is the logarithm of the average daily closing price during 1988, $LVAR$ is the logarithm of the daily return variance during 1988, and $LSIZE$ is the logarithm of the average daily market value of equity in 1988. $LNINST$ and $LPINST$ are logarithms of the number of institutions holding the stock and the percentage of shares held by institutions as of December 1987. IND_i is a dummy variable corresponding to one of five industry classifications, which are obtained using COMPUSTAT tapes and which follow Bhushan (1989).

The t -statistics are in parentheses.

Equation	(13)	(15)	(13)	(15)
Dependent				

(a) Table 2, part 1

372 M.J. Brennan, A. Subrahmanyam / Journal of Financial Economics 38 (1995) 361–381

Table 2 (continued)

Equation	(13)	(15)	(13)	(15)
Dependent variable	LTC_{GH}	$LVOL$	LTC_{MS}	$LVOL$
$LPRI$	0.275 (6.42)	-0.907 (15.80)	0.049 (0.92)	-0.919 (18.84)
$LVAR$	0.638 (18.40)		0.528 (12.23)	
$LSIZE$		0.615 (12.49)		0.673 (14.53)
LTC_{MS}				0.183

(b) Table 2, part 2

Table 2: Two-stage least-squares estimates of determinants of adverse-selection cost

Read:

- In the LTC_{GH} equation, the coefficient on $LANAL$ is -0.169 with a t-statistic of 3.13.
- In the LTC_{MS} equation, the coefficient on $LANAL$ is -0.258 with a t-statistic of 3.82.
- Trading volume has a strong negative relation with adverse-selection costs; volatility has a strong positive relation.

Economic message: After accounting for simultaneity, stocks with more analyst coverage have significantly lower adverse-selection costs. This supports the view that analyst competition and information production improve market depth.

9.3 Table 3: Robustness using the Foster-Viswanathan market-depth measure

corresponding values in Table 2, and remains negative and significant. It thus appears that our results are robust across these three different empirical approaches to estimating λ . As the theoretical models predict, the number of analysts following a stock has a significantly negative effect on the adverse selection cost of transacting in the stock.

Table 3

Two-stage least-squares estimates of determinants of adverse selection cost of transacting, using the Foster-Viswanathan empirical measure of λ :

$$LTC = a_{10} + a_{11}LANAL + a_{12}LVOL + a_{13}LPRI + a_{14}LVAR + e_{TC},$$

$$LANAL = a_{20} + a_{21}LTC + a_{22}LVAR + a_{23}LSIZE + a_{24}LPRI + \sum_{i=1}^5 a_{2i+4}IND_i + a_{210}LNINST + a_{211}LPINST + e_{ANAL},$$

$$LVOL = a_{30} + a_{31}LTC + a_{32}LANAL + a_{33}LSIZE + e_{VOL}.$$

The equation for $LANAL$ is not identified.

$LTC_{PV} = \log(\lambda_{PV}/PRI)$ is the log of the adverse selection costs of transacting. PRI is the average daily closing price.

λ_{PV} is the estimate of the inverse of market depth from the Foster-Viswanathan specification:

$$q_t = \alpha + \sum_{j=1}^5 \beta_j \Delta p_{t-j} + \sum_{k=1}^5 \gamma_k q_{t-k} + \tau_t, \quad \Delta p_t = \zeta + \lambda \tau_t + \psi[D_t - D_{t-1}] + \bar{u}_t.$$

Δp_t is the price change at transaction t , q_t is the signed trade size, and D_t is a dummy variable that is equal to +1 for a trade classified as a buy and -1 for a sell. The sample consists of 1,421 stocks continuously listed on the NYSE for the year 1988 for which complete data were available on institutional holdings as of December 1987 and for which nonnegative estimates of market depth were obtained.

The other variables are defined as follows: $LANAL$ is the logarithm of one plus the number of analysts as of December 1987, $LVOL$ is the logarithm of the average daily trading volume in 1988, $LPRI$ is the logarithm of the average daily closing price during 1988, $LVAR$ is the logarithm of the daily return variance during 1988, $LSIZE$ is the logarithm of the average daily market value of equity in 1988, $LNINST$ and $LPINST$ are logarithms of the number of institutions holding the stock and the percentage of shares held by institutions as of December 1987, IND_i is a dummy variable corresponding to one of five industry classifications, which are obtained using COMPUSTAT tapes and which follow Bhushan (1989).

The t -statistics are in parentheses.

M.J. Brennan, A. Subrahmanyam / Journal of Financial Economics 38 (1995) 361–381 375

Table 3 (continued)

Equation	(13)	(15)
Dependent variable	LTC_{PV}	$LVOL$
$LVOL$	-0.857 (24.59)	
$LPRI$	0.279 (6.36)	-0.920 (16.16)
$LVAR$	0.643 (18.16)	
$LSIZE$		0.630 (12.91)

(a) Table 3, part 1

(b) Table 3, part 2

Table 3: Two-stage least-squares estimates using the Foster-Viswanathan measure

Read:

- The coefficient on $LANAL$ in the adverse-selection cost equation is -0.185 with a t -statistic of 3.35 .
- The Foster-Viswanathan specification uses unexpected order flow rather than raw order flow.
- The results remain close to those from the Glosten-Harris and Madhavan-Smidt specifications.

Economic message: The main finding is not an artifact of a particular microstructure estimator. Analyst following continues to predict lower adverse-selection costs even when price impact is measured using unexpected order-flow innovations.

9.4 Table 4: Structural Admati-Pfleiderer model using trading-volume volatility

Table 4

Nonlinear two-stage least-squares estimates of two-equation Admati-Pfleiderer model using standard deviation of daily trading volume as proxy for standard deviation of noise trading:

$$LTC = a_0 + a_1 LINF + a_2 LSIGR + a_3 LSIGVOL + e_1,$$

$$\log(1+n) = b_0 + b_1 LTC + b_2 LSIGR + b_3 LPRI + k_1 LSIZE + \sum_{i=1}^5 IND_i + e_n.$$

LTC denotes the logarithm of the adverse selection cost of transacting, $\log(\lambda/PRI)$, where λ is estimated for a sample of 1,421 NYSE stocks using the Glosten-Harris and Madhavan-Smith methods and PRI is the average daily closing price in 1988. $LINF$ is defined as $LINF = \log[\sqrt{n/(n+1)}]$, where n is one plus the number of analysts as of December 1987. $LSIGR$ is the logarithm of the standard deviation of the rate of return in 1988, $LSIGVOL$ is the logarithm of the standard deviation of daily volume in 1988, $LPRI$ is the logarithm of the average daily closing price in 1988, $LSIZE$ is the logarithm of the average daily market value of equity in 1988. IND_i is a dummy variable corresponding to one of five industry classifications, which are obtained using COMPUSTAT tapes and which follow Bhushan (1989) (the coefficients on these variables are not reported for brevity).

The t -statistics in parentheses are asymptotic ones for testing whether the relevant coefficient is significantly different from zero. Theoretical values of the coefficients are shown in bold brackets.

Equation	(18)	(19)	(18)	(19)
Dependent variable	LTC_{GH}	$\log(1+n)$	LTC_{MS}	$\log(1+n)$
Constant	4.044 (5.19) [0.0]	-1.912 (6.59)	1.765 (1.42) [0.0]	-2.728 (7.47)
$LINF$	1.168 (1.87) [1.0]		4.276 (2.47) [1.0]	
LTC_{MS}				-0.358 (13.36) [-0.5]
LTC_{GH}		-0.307 (15.27) [-0.5]		
$LSIGR$	0.869 (12.08) [1.0]	0.431 (6.33) [1.0]	0.720 (9.39) [1.0]	0.467 (5.94) [1.0]
$LSIGVOL$	-1.028 (10.99) [-1.0]		-0.657 (7.46) [-1.0]	
$LPRI$		0.322 (7.39) [0.5]		0.210 (4.44) [0.5]
$LSIZE$		0.081 (4.11)		0.120 (5.88)

Table 4: Structural Admati-Pfleiderer estimates using standard deviation of daily trading volume

Read:

- $LINF$, the transformation of analyst following implied by the model, has the predicted positive sign in the LTC equation.
- $LSIGR$, return volatility, has a positive effect on adverse-selection cost, consistent with higher information uncertainty.
- $LSIGVOL$, the proxy for noise-trading volatility, has a negative coefficient, as predicted by the model.

Economic message: The structural estimates provide qualified support for the Admati-Pfleiderer mechanism: more informed competition reduces price impact, while greater fundamental uncertainty raises it.

9.5 Table 5: Structural Admati-Pfleiderer model using firm size as noise-trading proxy

things equal. In accordance with the role of $LSIZE$ as a proxy for noise trading, its coefficient is negative and significantly different from zero. Though the

Table 5

Nonlinear two-stage least-squares estimates of the first equation of the Admati-Pfleiderer model using firm size as a proxy for variance of noise trading:

$$LTC = a_0 + a_1 LINF + a_2 LSIGR + a_3 LSIZE + e_1,$$

$$\log(1 + n) = b_0 + b_1 LTC + b_2 LSIGR + b_3 LPRI + k_1 LSIZE + \sum_{i=1}^5 IND_i + e_n.$$

LTC denotes the logarithm of the adverse selection cost of transacting, $\log(\lambda/PRI)$, where λ is estimated for a sample of 1,421 NYSE stocks using the Glosten-Harris and Madhavan-Smidt methods and PRI is the average daily closing price in 1988. $LINF$ is defined as $LINF = \log[\sqrt{n/(n+1)}]$, as where n is one plus the number of analysts as of December 1987. $LSIGR$ is the logarithm of the standard deviation of the rate of return in 1988, $LPRI$ is the logarithm of the average daily closing price in 1988, $LSIZE$ is the logarithm of the average daily market value of equity in 1988. IND_i is a dummy variable corresponding to one of five industry classifications, which are obtained using COMPUSTAT tapes and which follow Bhushan (1989) (the coefficients on these variables are not reported for brevity).

Parameter estimates are given for only the first equation since the second is underidentified. The t -statistics in parentheses are asymptotic ones for testing whether the relevant coefficient is significantly different from zero. Theoretical values of the coefficients are shown in bold brackets.

	LTC_{GH}	LTC_{MS}
Constant	-4.033 (12.87) [0.0]	-5.309 (15.64) [0.0]
$LINF$	1.139 (2.90) [1.0]	2.208 (5.19) [1.0]
$LSIGR$	0.136 (1.69) [1.01]	0.321 (3.68) [1.01]

Table 5: Structural estimates using firm size as a proxy for noise trading

Read:

- The coefficient on $LINF$ is positive and statistically significant for both LTC_{GH} and LTC_{MS} .
- The coefficient on $LSIZE$ is negative and significant, consistent with larger firms having more noise trading and deeper markets.
- The model's first equation retains the predicted qualitative structure.

Economic message: When firm size is used as a proxy for noise trading, the analyst-following result strengthens. This suggests that the negative analyst-liquidity-cost relation is robust to alternative ways of measuring the uninformed trading base.

9.6 Table 6: Structural robustness with institutional ownership variables

Table 6

Nonlinear two-stage least-squares robustness test of two-equation Admati–Pfeiderer model using standard deviation of daily trading volume as a proxy for standard deviation of noise trading:

$$LTC = a_0 + a_1 LINF + a_2 LSIGR + a_3 LSIGVOL + e_1,$$

$$\log(1 + n) = b_0 + b_1 LTC + b_2 LSIGR + b_3 LPRI + b_4 LNINST$$

$$+ b_5 LPINST + k_1 LSIZE + \sum_{i=1}^5 IND_i + e_n.$$

LTC denotes the logarithm of the adverse selection cost of transacting, $\log(\lambda/PRI)$, where λ is estimated for a sample of 1,421 NYSE stocks using the Glosten–Harris and Madhavan–Smidt methods and PRI is the average daily closing price in 1988. $LINF$ is defined as $LINF = \log[\sqrt{n(n+1)}]$, where n is one plus the number of analysts as of December 1987. $LSIGR$ is the logarithm of the standard deviation of the rate of return in 1988, $LSIGVOL$ is the logarithm of the standard deviation of daily volume in 1988, $LPRI$ is the logarithm of the average daily closing price in 1988, $LNINST$ is the logarithm of number of institutions holding the stock as of December 1987, $LPINST$ is the logarithm of the percentage of the firm held by institutions as of December 1987, $LSIZE$ is the logarithm of the average daily market value of equity in 1988. IND_i is a dummy variable corresponding to one of five industry classifications, which are obtained using COMPUSTAT tapes and which follow Bhushan (1989) (the coefficients on these variables are not reported for brevity).

The t -statistics in parentheses are asymptotic ones for testing whether the relevant coefficient is significantly different from zero.

Equation	(18)	(19)	(18)	(19)
Dependent variable	LTC_{GH}	$\log(1 + n)$	LTC_{MS}	$\log(1 + n)$
<i>Constant</i>	4.785 (6.77)	− 0.701 (2.52)	3.402 (3.03)	− 0.858 (2.94)
<i>LINF</i>	1.594 (2.26)		3.820 (2.84)	
<i>LTC</i>		− 0.099 (4.80)		− 0.100 (4.78)
<i>LSIGR</i>	0.880 (16.57)	0.130 (2.14)	0.831 (13.63)	0.140 (2.26)
<i>LSIGVOL</i>	− 1.072 (15.49)		− 0.834 (11.35)	
<i>LPRI</i>		− 0.001 (0.02)		− 0.034 (0.81)
<i>LSIZE</i>		0.017 (0.38)		0.032 (0.73)
<i>LNINST</i>		0.464 (6.78)		0.474 (6.84)
<i>LPINST</i>		0.119		0.177

Table 6: Structural robustness test adding institutional ownership variables

Read:

- Adding institutional ownership variables does not materially change the adverse-selection cost equation.
- The number of institutions and the percentage held by institutions significantly explain analyst following.
- The R^2 for the analyst-following equation rises substantially relative to the simpler structural specification.

Economic message: Analyst coverage is shaped not only by the theoretical costs and benefits of becoming informed, but also by institutional demand for information. The enriched specification fits the market for information better.

10 Interpretation and implications

10.1 For market microstructure

The paper shows that information production can improve market liquidity, not only harm it. Although informed traders create adverse selection for market makers, competition among informed

traders can reduce informational rents and deepen the market.

10.2 For analyst coverage

Analyst coverage has a market-level function beyond forecasting earnings. It appears to improve the price formation process by reducing adverse-selection costs. This provides a channel through which financial intermediaries and information producers can create social value.

10.3 For asset pricing

Since market illiquidity can affect required returns, the paper implies that analyst following may matter for expected returns through a liquidity channel. Neglected firms may face higher trading costs and potentially higher required returns.

11 Limitations

- Analyst following is an imperfect proxy for informed traders; analysts produce information, but actual trading may be conducted by institutions or clients.
- The sample is restricted to NYSE stocks in 1988, so the results may not directly generalize to later electronic markets.
- Estimating adverse-selection costs from intraday data requires assumptions about trade signing, order-flow exogeneity, and the price impact model.
- The structural Admati-Pfleiderer model is highly stylized and cannot fully capture institutional features of analyst coverage.

12 Connection to other papers

- **Kyle (1985):** provides the market-depth framework where price impact reflects information in order flow.
- **Admati and Pfleiderer (1988):** predicts that competition among informed traders can increase depth.
- **Glosten and Harris (1988):** provides an empirical way to estimate adverse-selection components of the spread.
- **Madhavan and Smidt (1991):** allows order flow to condition on prices and provides another price-impact estimator.
- **Bhushan (1989):** motivates institutional ownership and firm characteristics as determinants of analyst following.

13 Short summary

Brennan and Subrahmanyam (1995) show that analyst coverage is negatively associated with adverse-selection trading costs in NYSE stocks. Using multiple intraday price-impact estimators and simultaneous-equation regressions, the paper finds that more analysts following a stock imply lower inverse market depth, after controlling for trading volume, price, volatility, size, and

institutional ownership. The result supports the view that competition in information production improves price formation and market liquidity.

14 Conclusion

The central lesson is that securities analysis affects how prices form in markets. More analyst following appears to make markets deeper by reducing the price impact of trades. The paper therefore connects three literatures: information acquisition, market microstructure, and liquidity-based asset pricing. Its main empirical result is robust across alternative estimates of adverse-selection costs and is broadly consistent with the Admati-Pfleiderer model of informed-trader competition.